# Exact unrestricted Gaussian rank of two canonical matchgate magic states

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Gaussian rank is the smallest number of pure fermionic Gaussian states whose complex linear combination equals a target state, and it controls the branch count in rank-based simulations of matchgate circuits with non-Gaussian inputs. For the canonical four-mode state $|M\rangle = (|0000\rangle + |1111\rangle)/\sqrt{2}$, previous work established rank four for $|M\rangle^{\otimes 2}$ only when every summand obeys two additional blockwise symmetries, and conjectured the same value without those restrictions. We prove the unrestricted statement over $\mathbb{C}$. After a parity reduction, a nonzero Chevalley pairing places any hypothetical three-Gaussian decomposition, up to $\mathrm{Spin}(16,\mathbb{C})$, in one of 15 normal forms $a\,1+b\,\Omega+c\,g_{q,i}$. A two-form Clifford flattening has exact rank 78 on the target. Exact characteristic-zero certificates put the 15 normal forms on the two sides of a strict rank gap: eleven have rank at most 74 and four have rank at least 83. The certificate is checked by two separately implemented exterior-algebra calculations. Together with the four computational-basis Gaussian summands, this gives $\chi_G(|M\rangle^{\otimes 2}) = 4$ with no reality, particle-number, copy-exchange, block-symmetry, or Pfaffian-chart assumption. The result concerns exact rank for this finite eight-mode state; it does not determine border rank, approximate rank, or many-copy asymptotics.

## I. INTRODUCTION

Matchgate circuits and fermionic linear optics form a nontrivial class of quantum processes admitting efficient classical descriptions [1–3]. Supplementing them with non-Gaussian resources changes their computational power [4, 5]. One way to simulate such a supplemented circuit is to decompose its input into pure fermionic Gaussian states and propagate each Gaussian branch. The minimum exact number of branches is the Gaussian rank.

Cudby and Strelchuk introduced and studied this rank-based viewpoint for matchgate magic states [6]. For their canonical four-mode state $|M\rangle$, they displayed the immediate four-term Gaussian decomposition of two copies. They proved the matching lower bound when each Gaussian summand is constrained to the $+1$ eigenspaces of two blockwise $ZZ$ operators. Their numerical search found no unrestricted decomposition with three terms, leading to the conjecture

$$\chi_G(|M\rangle^{\otimes 2}) = 4. \tag{1}$$

The symmetry-restricted lower bound does not decide Eq. (1), because individual terms in a decomposition need not inherit a symmetry of the sum.

Here we resolve Eq. (1) without restrictions on the Gaussian summands. The proof combines a complete pure-spinor normal-form reduction with an exact Clifford-flattening certificate. Secant varieties of spinor varieties have been studied at the level of dimensions and defectivity [7]; those global dimensions do not by themselves decide whether this particular target lies on the affine third secant. The target-specific rank gap below does.

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The argument is organized as follows. Section II fixes the fermionic conventions and the exact-rank question. Section III reduces every possible three-term decomposition to 15 normal forms. Sections IV and V construct and certify the rank gap. Section VI records the computer-assisted component and error control. We close with the operational meaning and explicit scope limitations.

## II. EIGHT-MODE TARGET AND GAUSSIAN RANK

Let $\mathcal{H}_8 = (\mathbb{C}^2)^{\otimes 8}$ be the Fock space of eight ordered fermionic modes. We use the Jordan–Wigner Majoranas

$$c_{2j-1} = Z_1 \cdots Z_{j-1} X_j, \qquad c_{2j} = Z_1 \cdots Z_{j-1} Y_j, \tag{2}$$

for $j = 1, \ldots, 8$, so that $\{c_r, c_s\} = 2\delta_{rs} I$. Define

$$\Lambda_8 = \sum_{r=1}^{16} c_r \otimes c_r. \tag{3}$$

In the intrinsic criterion used in Ref. [6], a normalized pure state is fermionic Gaussian precisely when

$$\Lambda_8(|g\rangle \otimes |g\rangle) = 0. \tag{4}$$

This definition allows either parity and does not assume particle-number conservation.

For a nonzero state $|\phi\rangle$, its exact Gaussian rank over $\mathbb{C}$ is

$$\chi_G(\phi) = \min\Big\{ r : |\phi\rangle = \sum_{k=1}^{r} \alpha_k |g_k\rangle, \quad \alpha_k \in \mathbb{C}, \quad |g_k\rangle \in \mathcal{G}_8 \Big\}. \tag{5}$$

Normalization of each $|g_k\rangle$ causes no loss, since its scale can be absorbed into $\alpha_k$.

The four-mode and eight-mode states of interest are

$$|M\rangle = \frac{|0000\rangle + |1111\rangle}{\sqrt{2}}, \tag{6}$$

$$|\Psi\rangle = |M\rangle^{\otimes 2} = \frac{1}{2}(|0\rangle + |15\rangle + |240\rangle + |255\rangle), \tag{7}$$

where the decimal labels denote the corresponding eight-bit Fock strings. Every computational-basis vector is a pure Gaussian state, and direct substitution in Eq. (4) gives zero. Equation (7) therefore proves $\chi_G(\Psi) \leq 4$. We work below with the unnormalized vector

$$x = 2|\Psi\rangle = |0\rangle + |15\rangle + |240\rangle + |255\rangle, \tag{8}$$

which has the same rank.

## III. COMPLETE REDUCTION OF THREE GAUSSIAN SUMMANDS

Set $E = \mathbb{C}^8$ and use the exterior model $S^+ = \Lambda^{\text{even}} E$ for the positive half-spin representation. Nonzero even Gaussian vectors are exactly the affine cone over the even pure-spinor variety [8]. Creation and annihilation operators act as exterior multiplication $\epsilon_j$ and contraction $\iota_j$.

### A. Parity reduction

Pure fermionic Gaussian vectors have definite parity. If the even nonzero vector $x$ were a sum of at most three definite-parity Gaussian vectors, its odd projection would have to vanish. One odd term must therefore have zero coefficient; two odd terms must be proportional and cancel; and three odd terms would leave no nonzero even part. In the two-odd-term case the remaining even part would be a single pure spinor. More generally, the flattening introduced in Sec. IV has rank 29 on each pure spinor and rank 78 on $x$, so neither one nor two pure spinors can sum to $x$. It is therefore sufficient to exclude sums of three even pure spinors.

### B. A transverse pair

Let $\Omega = e_1 \wedge \cdots \wedge e_8$ and define the invariant Chevalley pairing by

$$B(s,t) = [\tau(s) \wedge t]_{\Lambda^8 E}, \tag{9}$$

where $\tau$ reverses wedge order. A pure spinor pairs to zero with itself. Direct evaluation of the four components in Eq. (8) gives

$$B(x,x) = 4 \neq 0. \tag{10}$$

Consequently, in any equality $x = s_1 + s_2 + s_3$ of pure spinors, at least one pair $s_j, s_k$ has nonzero pairing.

For pure spinors, nonzero Chevalley pairing is equivalent to transversality of their maximal-isotropic annihilators. The complex spin group acts transitively on ordered transverse pairs, taking their projective lines to those of 1 and $\Omega$. The common projective stabilizer is the lifted $GL(E)$. Hence a hypothetical three-term decomposition can be transformed into

$$u = a\,1 + b\,\Omega + c\,g, \qquad a, b, c \in \mathbb{C}^*, \tag{11}$$

with $g$ an even pure spinor. The coefficients are nonzero: if any vanished, the flattening subadditivity bound would give rank at most $2 \times 29 = 58$, in conflict with the target rank 78.

### C. Fifteen GL(8) orbits

For completeness, the required pure-spinor normal form can be obtained directly from its annihilator equations. If $q$ is the lowest nonzero exterior degree of $g$, its degree-$q$ component is decomposable. A change of basis makes it $e_1 \wedge \cdots \wedge e_q$, and purity recursively gives

$$g = e_1 \wedge \cdots \wedge e_q \wedge \exp(\omega), \qquad \omega \in \Lambda^2 \operatorname{span}(e_{q+1}, \ldots, e_8). \tag{12}$$

Because $g \in S^+$, $q \in \{0, 2, 4, 6, 8\}$. Complex skew congruence reduces $\omega$ to a Darboux form of rank $2i$. Absorbing the projective scale into $c$ leaves

$$g_{q,i} = e_1 \cdots e_q \prod_{j=1}^{i} (1 + e_{q+2j-1} e_{q+2j}), \quad 0 \leq i \leq \frac{8-q}{2}. \tag{13}$$

Thus $5 + 4 + 3 + 2 + 1 = 15$ cases exhaust all even pure spinors, including those with zero vacuum amplitude. No real structure or target symmetry has entered this reduction.

## IV. TWO-FORM CLIFFORD FLATTENING

Let $V = E \oplus E^*$ be the 16-dimensional quadratic space and let $\Gamma_1, \ldots, \Gamma_{16}$ be split Clifford generators. For $u \in S^+$, define

$$F_u : S^+ \longrightarrow \Lambda^2 V^*, \tag{14}$$

$$[F_u(y)]_{rs} = B(u, \Gamma_r \Gamma_s y), \qquad r < s. \tag{15}$$

This is a $120 \times 128$ matrix in the natural exterior bases. Invariance of $B$ and covariance of Clifford multiplication imply

$$\operatorname{rank} F_{h \cdot u} = \operatorname{rank} F_u, \qquad h \in \operatorname{Spin}(16, \mathbb{C}). \tag{16}$$

The physical Majoranas in Eq. (2) correspond to $\epsilon_j + \iota_j$ and $i(\epsilon_j - \iota_j)$. Relative to the split generators they only multiply rows of $F_u$ by nonzero powers of $i$, so the rank is unchanged.

For the vacuum, only the degree-eight and degree-six columns supply complementary top forms in Eq. (15). Direct basis action gives

$$\operatorname{rank} F_1 = 1 + \binom{8}{2} = 29. \tag{17}$$

Equivariance then gives rank 29 for every nonzero even pure spinor. This also justifies the rank-at-most-58 assertions used above.

The flattening is used here as an orbit-separating covariant [9], not as the usual subadditive lower bound alone. Indeed, the generic bound from three pure terms would only be $3 \times 29 = 87$, whereas the target rank is 78. The decisive fact is that the complete normal forms in Eq. (13) avoid rank 78.

## V. EXACT CHARACTERISTIC-ZERO RANK-GAP CERTIFICATE

Rows are ordered lexicographically by the 120 pairs $r < s$, and columns by the 128 even occupation masks. All entries of $F_x$ are integers. The certificate exhibits a $78 \times 78$ minor with

$$\det = 2361183241434822606848 = 2^{71}, \tag{18}$$

together with 50 linearly independent right-null vectors. These give matching lower and upper bounds,

$$\operatorname{rank} F_x = 78. \tag{19}$$

For each normal form

$$u_{q,i}(a,b,c) = a\,1 + b\,\Omega + c\,g_{q,i}, \tag{20}$$

the same matrix is treated over the rational-function field $\mathbb{Q}(a,b,c)$. Table I summarizes the certified alternatives.

For the eleven low cases, the certificate supplies 71 independent rational right-null vectors when the upper bound is 57, and 54 when it is 74. A possible denominator zero creates no exceptional loophole: the nullspace identities over $\mathbb{Q}(a,b,c)$ imply that every 58- or 75-minor, respectively, is the zero polynomial. Matrix rank cannot increase under specialization.

For the four high cases, the nonzero pattern separates into seven bipartite row-column support components. Selecting a minor in each component gives disjoint rows and columns, so the minor sizes add. The bounds are 84 for $(0,3)$, 83 for $(0,4)$, 83 for $(2,2)$, and 84 for $(2,3)$. All selected determinants are nonzero monomials in $a,b,c$ except for the first component of $(0,4)$. There the certificate provides two alternative $11 \times 11$ minors,

$$-128a^2b^3c^5(a-b), \qquad 64a^2b^3c^5(a+b). \tag{21}$$

For $abc \neq 0$ in characteristic zero, $a-b$ and $a+b$ cannot vanish simultaneously. Thus the bound 83 holds on both exceptional loci as well as generically.

We may now state the result.

*Theorem.* For $|M\rangle = (|0000\rangle + |1111\rangle)/\sqrt{2}$, the exact unrestricted fermionic Gaussian rank over $\mathbb{C}$ is

$$\boxed{\chi_G(|M\rangle^{\otimes 2}) = 4}. \tag{22}$$

*Proof.* The four Fock-basis components in Eq. (7) are Gaussian and give rank at most four. Suppose the rank were at most three. Parity reduces the putative decomposition to three even pure spinors. Equations (10)–(13) transform it, without changing flattening rank, to one of the 15 matrices in Eq. (20). Equation (19) requires rank 78. Table I and the exceptional certificates (21) instead give rank at most 74 or at least 83 in every case, a contradiction. Therefore the rank is at least four, which meets the upper bound. □

## VI. COMPUTER-ASSISTED VERIFICATION AND ERROR CONTROL

The computer-assisted part is a finite exact certificate, not a numerical scan. The generator constructs $F_u$ from bit-mask wedge and contraction operations, computes the target pivot minor and nullspace, obtains rational-function nullspaces for all low strata, and verifies every indexed high-stratum minor before emitting a machine-readable record. A second implementation reconstructs the matrices through separately written wedge, contraction, and top-wedge reversal operations. It reads only the claimed certificate indices and vectors, then recomputes every dot product, determinant, support component, and rank. Both implementations use exact integer and rational-function arithmetic in SymPy [10].

OpenAI Codex workers assisted with certificate design and code generation. The registered model identifier was `gpt-5.6-sol`, with reasoning effort `ultra`. They were directed to reproduce the baseline, restrict the claim to exact eight-mode rank, and require a separate replay. Conclusions rest on the retained exact identities, not model confidence; human inspection remains required.

The replay additionally substitutes directly on $a = b$ and $a = -b$ for the $(0,4)$ stratum and obtains exact ranks 83 and 84, respectively. It checks all 15 strata and reconstructs the target from the two four-mode factor supports. The submitted certificate and scripts provide the row/column indices and sparse null vectors; thus no trust in a reported rank routine is needed to check the central identities.

There is no floating-point tolerance, statistical sampling, truncation, fitted degree bound, finite-field lifting, or perturbative remainder. A prior finite-field scan was used only to discover the rank pattern and is not part of the proof. The analytic inputs that remain for human inspection are the parity, transverse-pair, and pure-spinor normal-form reductions; the exact replay checks the finite matrices conditional on those reductions.

TABLE I. Exact bounds for all 15 normal forms with $abc \neq 0$. An upper bound over $\mathbb{Q}(a, b, c)$ holds after every complex specialization because all larger minors vanish identically. Each lower bound is supplied by mutually row- and column-disjoint nonzero minors.

| Type | $(q, i)$ | Certified statement | Separation from rank $F_x = 78$ |
|---|---|---|---|
| Low | $(0,0), (0,1), (2,0), (6,0), (6,1), (8,0)$ | rank $F_{u_{q,i}} \leq 57$ | at most 74 |
| Low | $(0,2), (2,1), (4,0), (4,1), (4,2)$ | rank $F_{u_{q,i}} \leq 74$ | at most 74 |
| High | $(0,3), (2,3)$ | rank $F_{u_{q,i}} \geq 84$ | at least 83 |
| High | $(0,4), (2,2)$ | rank $F_{u_{q,i}} \geq 83$ | at least 83 |

## VII. PHYSICAL MEANING AND LIMITATIONS

The theorem closes the gap between the known four-branch decomposition and the previous lower bound restricted by two blockwise $ZZ$ symmetries [6]. It shows that symmetry breaking, arbitrary complex phases, non-number-conserving Gaussian states, and zero-vacuum-amplitude charts cannot reduce the exact branch count below four for this two-copy input. In a simulation that propagates an exact Gaussian decomposition branch by branch, the input therefore requires exactly four Gaussian branches.

This operational statement is deliberately finite. Exact rank is not border rank: the proof neither excludes a sequence of three-term states converging to $|\Psi\rangle$ nor supplies an error-versus-cost bound for approximate simulation. It also does not determine $\chi_G(|M\rangle^{\otimes k})$ for $k \geq 3$, a regularized rank, or an asymptotic runtime exponent. The result is a property of the specified eight-mode state and Gaussian resource set, not evidence for a new particle, phase, or experimental effect.

## VIII. CONCLUSION

An unrestricted rank-three decomposition of two canonical matchgate magic states would, after an intrinsic pure-spinor reduction, have to occupy one of only 15 $GL(8)$ normal forms. The two-form Clifford flattening places the target at rank 78 and every such form outside the interval from 75 through 82. This exact rank gap proves Gaussian rank four over $\mathbb{C}$ and removes all auxiliary symmetries from the earlier two-copy lower bound. The same strategy—orbit reduction followed by a covariant rank gap—may be useful for other target-specific secant-membership questions, although no many-copy claim follows from the present calculation.

## ACKNOWLEDGMENTS

In addition to the research uses disclosed in Sec. VI, OpenAI Codex assisted with literature synthesis and drafting. The tool is not an author, and this draft requires the named author's independent review and refinement before submission.

## DATA AVAILABILITY

The exact certificate and the two Python programs needed to regenerate and replay the computer-assisted statements accompany this manuscript. No experimental or personal data were used.

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